

Problem

For the following pairs of signals, determine if v_1 leads or lags v_2 and by how much.

(a) $v_1 = 10 \cos(5t - 20^\circ)$, $v_2 = 8 \sin 5t$

(b) $v_1 = 19 \cos(2t + 90^\circ)$, $v_2 = 6 \sin 2t$

(c) $v_1 = -4 \cos 10t$, $v_2 = 15 \sin 10t$

Step-by-step solution

Step 1 of 3

(a)

Consider the following pair of signals.

$$v_1 = 10 \cos(5t - 20^\circ)$$

$$v_2 = 8 \sin 5t$$

Represent the signal v_2 in cosine form.

$$v_2 = 8 \sin 5t$$

$$= 8 \cos(90^\circ - 5t) \quad \left(\text{since, } \cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta \right)$$

$$= 8 \cos(5t - 90^\circ) \quad \left(\text{since, } \cos(-\theta) = \cos \theta \right)$$

Compare the expressions for the signals v_1 and v_2 to get the following.

The v_1 signal **leads** the v_2 signal by 70° .

Step 2 of 3

(b)

Consider the following pair of signals.

$$v_1 = 19 \cos(2t + 90^\circ)$$

$$v_2 = 6 \sin 2t$$

Represent the signal v_2 in cosine form.

$$v_2 = 6 \sin 2t$$

$$= 6 \cos(90^\circ - 2t) \quad \left(\text{since, } \cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta \right)$$

$$= 6 \cos(2t - 90^\circ) \quad \left(\text{since, } \cos(-\theta) = \cos \theta \right)$$

Compare the expressions for the signals v_1 and v_2 to get the following.

The v_1 signal **leads** the v_2 signal by 180° .

Step 3 of 3

(c)

Consider the following pair of signals.

$$v_1 = -4 \cos 10t$$

$$v_2 = 15 \sin 10t$$

Represent the signal v_1 in general form.

$$v_1 = -4 \cos 10t$$

$$= 4 \cos(180^\circ + 10t) \quad \left(\text{since, } \cos(\pi + \theta) = -\cos \theta \right)$$

$$= 4 \cos(10t + 180^\circ)$$

Represent the signal v_2 in cosine form.

$$v_2 = 15 \sin 10t$$

$$= 15 \cos(90^\circ - 10t) \quad \left(\text{since, } \cos\left(\frac{\pi}{2} - \theta\right) = \sin \theta \right)$$

$$= 15 \cos(10t - 90^\circ) \quad \left(\text{since, } \cos(-\theta) = \cos \theta \right)$$

Compare the expressions for the signals v_1 and v_2 to get the following.

The v_1 signal **lags** the v_2 signal by 270° .